Company Bankruptcy Prediction

Predicting corporate bankruptcy from financial ratios with XGBoost

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1. Data Overview

The dataset contains financial information on companies, where each row is one company and each column is a financial ratio or indicator. The target variable **Bankrupt?** shows whether the company went bankrupt (1) or not (0). The goal of this project is to build a classifier that identifies companies at risk of bankruptcy.

1.1 Data Dimensions and Structure

Table 1. Dataset dimensions

Item	Value
Number of rows (companies)	6,819
Number of columns	96 (95 features + 1 target)
Float columns	93
Integer columns	3 (Bankrupt?, Liability-Assets Flag, Net Income Flag)
Target variable	Bankrupt? (0 = not bankrupt, 1 = bankrupt)

All features are numeric. Most are financial ratios covering profitability (e.g. ROA, profit margins), leverage (e.g. debt ratio, liability to equity), liquidity (e.g. current ratio, quick ratio), cash flow, turnover and growth rates. The column names in the raw file contain leading spaces, so they were stripped before analysis.

1.2 Data Quality Check

Table 2. Data quality summary

Check	Result
Missing values	0 in all columns
Duplicate rows	0
Infinite values	0
Constant columns	1 (Net Income Flag, which has the same value for every company)
Rows with at least one IQR outlier	6,361 of 6,819 (93.3%), including 219 of the 220 bankrupt companies

The data is complete and clean in the sense of missing and duplicated values. The main quality issues are the constant column, which carries no information, and the large number of outliers.

1.3 Target Variable Distribution

Table 3. Target class balance

Class	Count	Percent
Not bankrupt (0)	6,599	96.77%
Bankrupt (1)	220	3.23%

The dataset is **highly imbalanced**: only 3.23% of the companies are bankrupt, a ratio of about 30 to 1. This has two important consequences. First, a model that always predicts "not bankrupt" already achieves 96.8% accuracy, so accuracy on its own is not a meaningful metric. Second, a model trained on this data tends to favour the majority class, so special handling (resampling, class weights or threshold tuning) is needed to detect bankrupt companies.

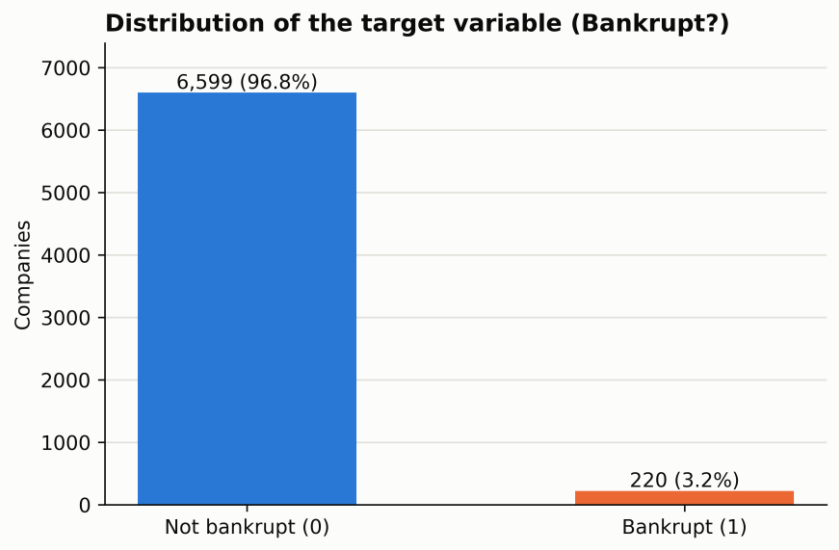


Figure 1.1. Distribution of the target variable

1.4 Outliers and Feature Scale

Outliers were counted with the IQR rule: a value is an outlier if it lies below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Several features have more than 20% of rows flagged, led by Degree of Financial Leverage (22.0%), Interest Coverage Ratio (20.8%) and Fixed Assets Turnover Frequency (20.8%). Because 93.3% of all rows contain at least one outlier on some feature, removing every outlier row would delete almost the entire dataset.

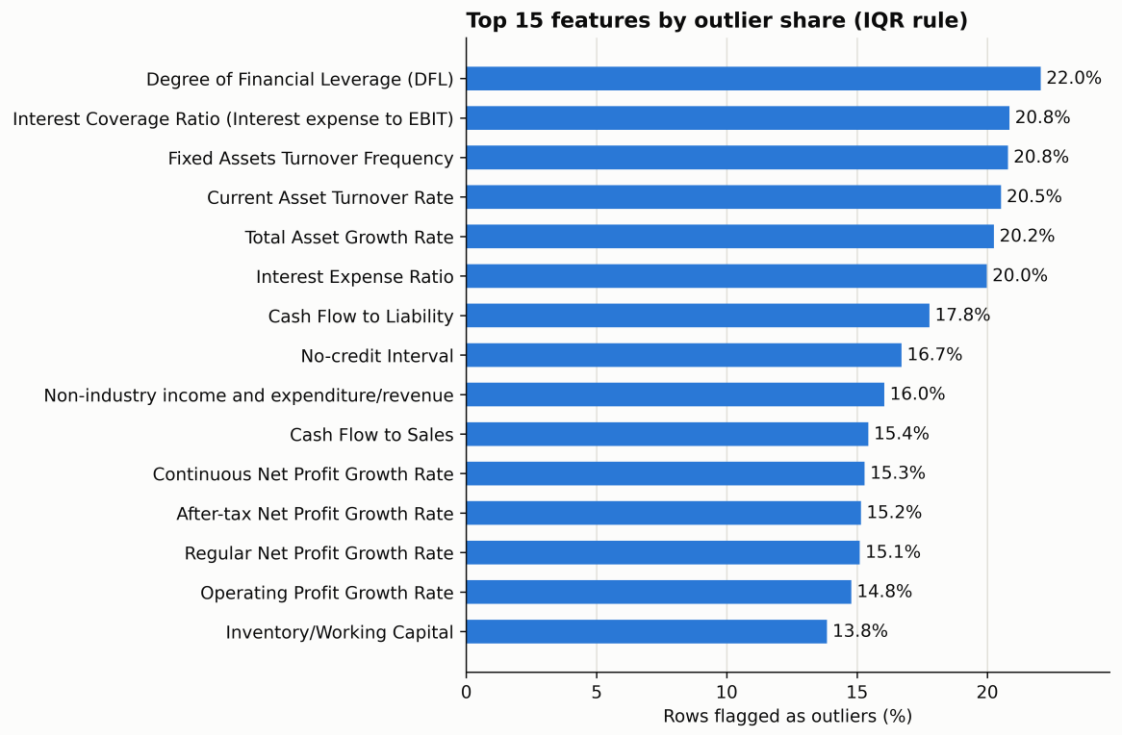


Figure 1.2. Top 15 features by share of rows flagged as outliers (IQR rule)

The features are on very different scales. 71 of the 95 features lie between 0 and 1, which suggests they were normalised in the source data, while 24 features have maximum values above one million, and 23 of these exceed one billion. These extreme values are most likely recording or scaling errors. Tree-based models such as XGBoost split on thresholds, so they are not sensitive to feature scale, and no scaling was applied.

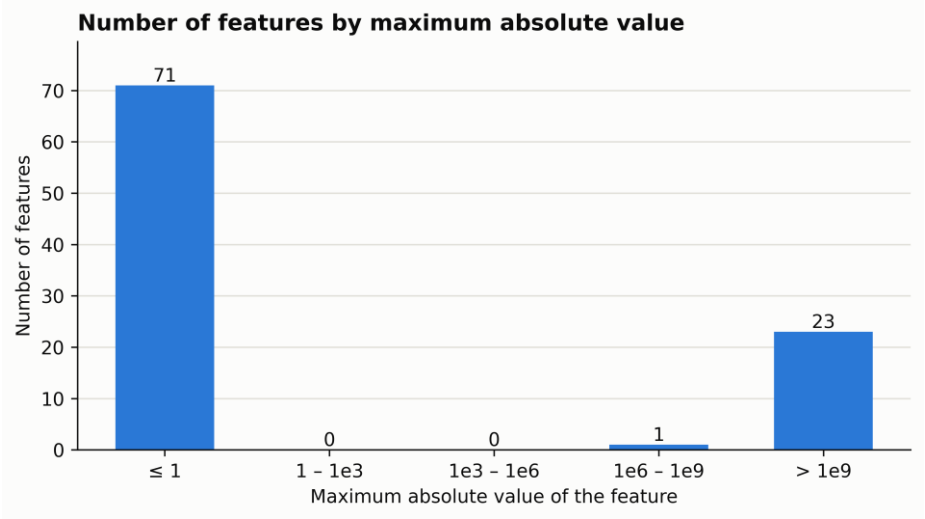


Figure 1.3. Number of features by maximum absolute value

2. Exploratory Data Analysis (EDA)

2.1 Descriptive Statistics

Summary statistics (count, mean, standard deviation, quartiles, minimum, maximum, skewness and kurtosis) were computed for all 95 features. The distributions are strongly non-normal: **81 of 95 features** have an absolute skewness above 1, and **48 features** have an absolute skewness above 10. This is consistent with the heavy outliers found in Section 1.4 and supports the choice of a tree-based model, which does not assume normally distributed inputs.

2.2 Feature–Target Relationship

The Pearson correlation between each feature and the target was calculated (the constant column was excluded). The strongest relationships are all fairly weak, with the largest absolute correlation at 0.315.

Table 4. Top 10 features by absolute correlation with the target

Feature	Correlation with Bankrupt?
Net Income to Total Assets	−0.315
ROA(A) before interest and % after tax	−0.283
ROA(B) before interest and depreciation after tax	−0.273
ROA(C) before interest and depreciation before interest	−0.261
Debt ratio %	+0.250
Net worth/Assets	−0.250
Persistent EPS in the Last Four Seasons	−0.220
Retained Earnings to Total Assets	−0.218
Net profit before tax/Paid-in capital	−0.208
Per Share Net profit before tax (Yuan ¥)	−0.201

The results make financial sense. **Profitability** ratios (net income to assets, ROA, EPS, retained earnings) are negatively correlated with bankruptcy, so less profitable companies are more likely to fail. **Leverage** ratios (debt ratio, current liability to assets, borrowing dependency) are positively correlated, so companies that rely more on debt are at higher risk.

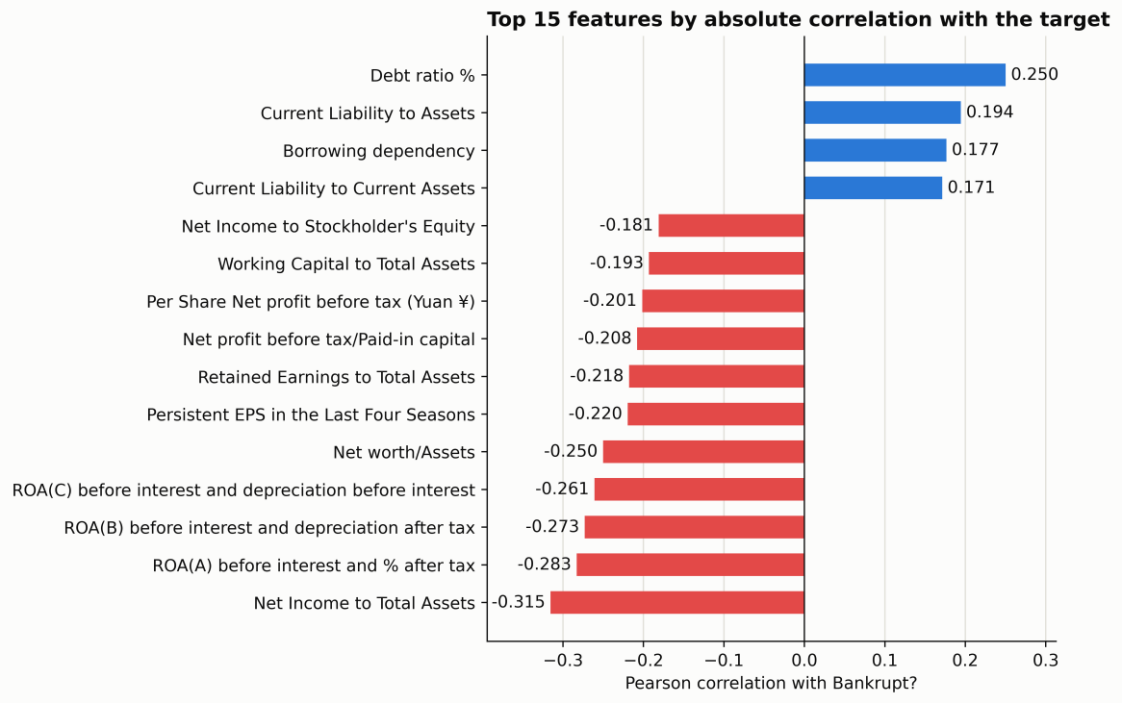


Figure 2.1. Top 15 features by absolute correlation with the target

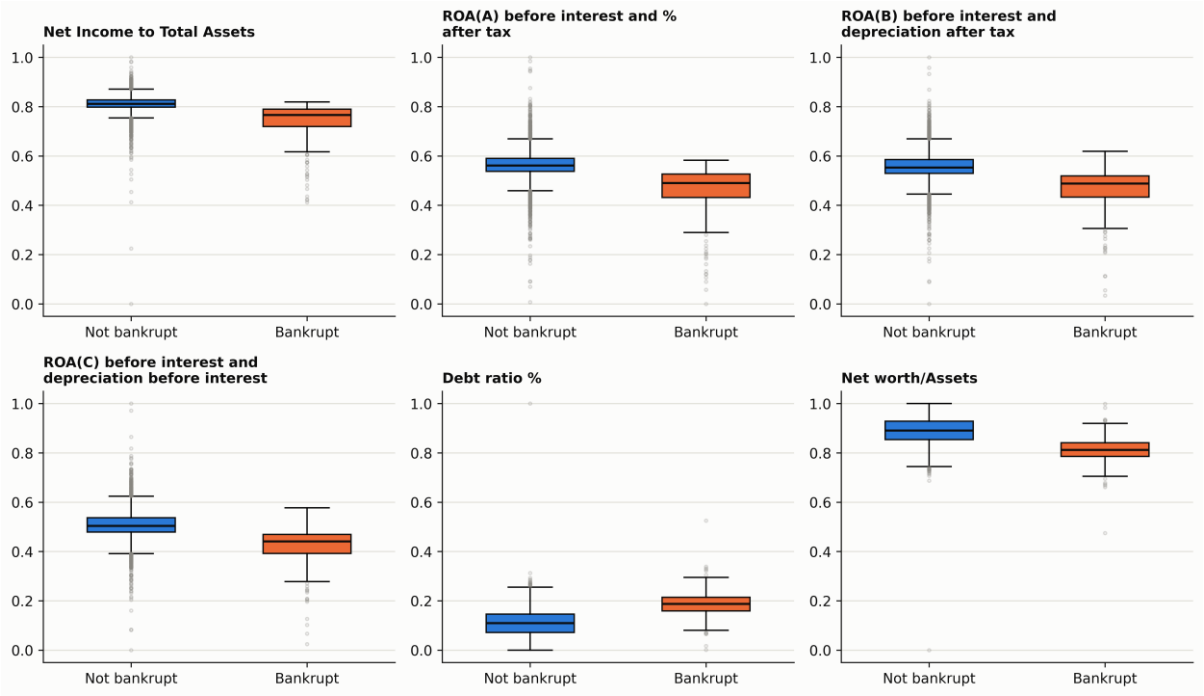


Figure 2.2. Distribution of the six most correlated features by class

The box plots in Figure 2.2 confirm the pattern. Bankrupt companies have clearly lower profitability ratios and a higher debt ratio than healthy companies, although the two classes still overlap a lot.

2.3 Correlation Between Features (Multicollinearity)

Many features measure almost the same thing. There are **28 feature pairs with |r| > 0.90** and **23 pairs with |r| > 0.95**. Several pairs are perfectly correlated (|r| = 1.00), for example:

- Current Liabilities/Equity and Current Liability to Equity

- Current Liabilities/Liability and Current Liability to Liability
- Debt ratio % and Net worth/Assets
- Operating Gross Margin and Gross Profit to Sales

The three ROA variants and the three Net Value Per Share variants are also correlated at above 0.98. These duplicated features add no new information and make correlation-based feature selection unreliable. This is why the engineered model selects features by XGBoost importance instead (Section 3).

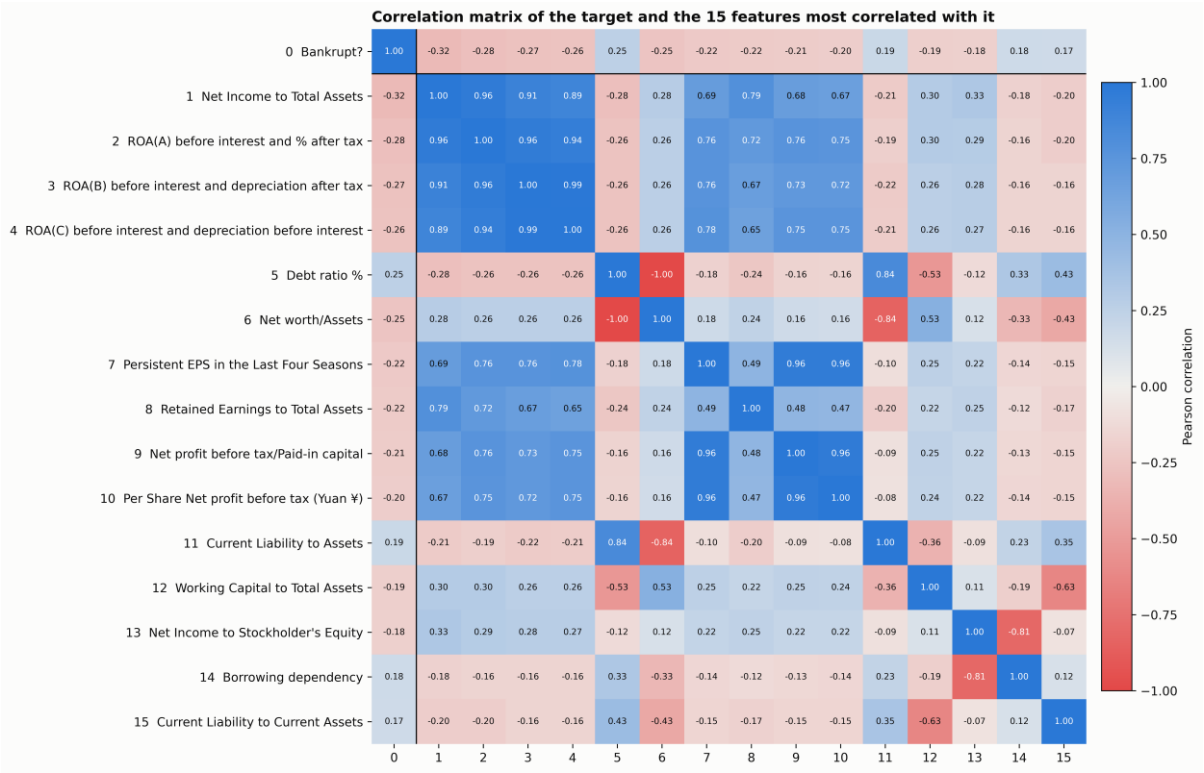


Figure 2.3. Correlation matrix of the target (row/column 0) and the 15 features most correlated with it

2.4 Feature Distributions by Target

Figure 2.4 compares the distributions of the 12 most correlated features for the two classes. Densities are calculated separately for each class, so the 220 bankrupt companies can be compared fairly with the 6,599 others, and only the central 99% of values is shown so that extreme outliers do not squash the plot. Bankrupt companies are shifted towards lower profitability and higher liabilities, but no single feature separates the classes cleanly. A multivariate model is therefore needed.

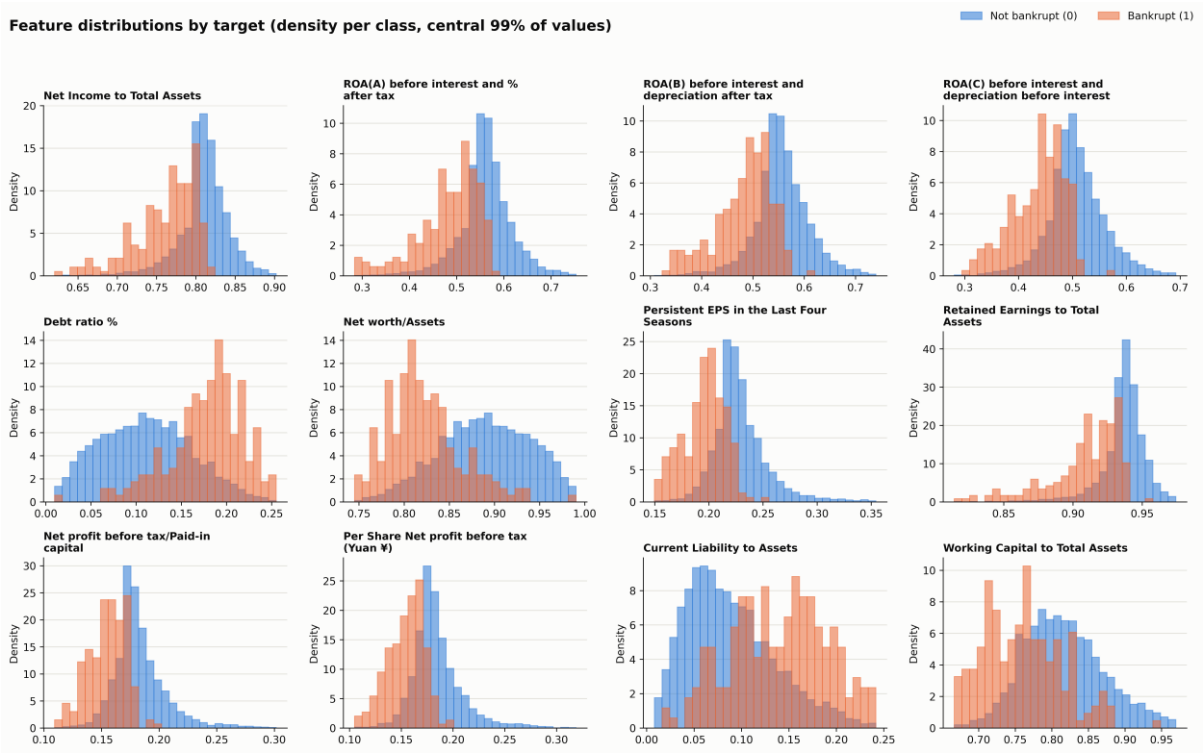


Figure 2.4. Feature distributions by target (density per class)

2.5 Number of Features Correlated with the Target

Table 5. Features by strength of linear correlation with the target

|r| with target	Positive	Negative	Total
< 0.05	13	32	45
0.05 – 0.10	6	11	17
0.10 – 0.20	9	13	22
0.20 – 0.30	1	8	9
≥ 0.30	0	1	1
All features	**29**	**65**	**94**

Almost half of the features (45 of 94) have an absolute correlation below 0.05, and only 10 features reach 0.20 or more. Most features are negatively correlated with bankruptcy. The weak linear correlations do not mean the features are useless, because non-linear effects and interactions can still be captured by a tree-based model.

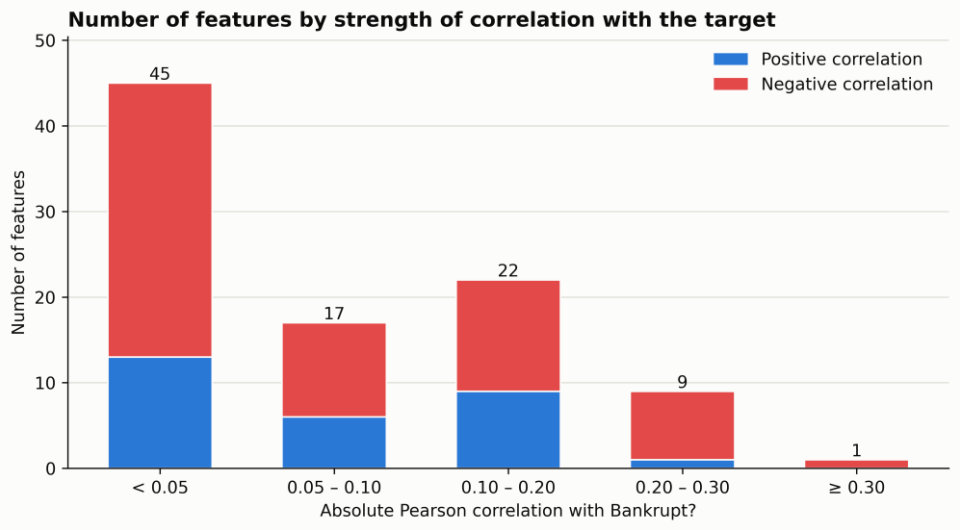


Figure 2.5. Number of features by strength of correlation with the target

3. Data Pre-processing

3.1 Cleaning

- Leading spaces were stripped from all column names.
- No imputation or deduplication was needed (0 missing values, 0 duplicates, 0 infinite values).
- No feature scaling was applied, because XGBoost is not sensitive to feature scale.
- The constant column (Net Income Flag) was kept in the data. It has no variance, so XGBoost never splits on it.

3.2 Train–Test Split and Cross-Validation

The data was split **before** any other processing into an 80% training set and a 20% test set, stratified on the target so that both sets keep the 3.2% bankruptcy rate. The test set was not used until the final comparison.

Table 6. Train–test split (stratified, random_state = 42)

Set	Rows	Bankrupt companies
Training	5,455	176
Test	1,364	44

Model evaluation and tuning on the training set used **stratified 10-fold cross-validation**.

3.3 Pre-processing Steps for the Engineered Model

For the engineered model, all pre-processing steps sit inside one imbalanced-learn Pipeline. Within each cross-validation fold, the steps are fitted on that fold's training part only, and the resampling steps (outlier removal and SMOTE) are never applied to validation or test data. This prevents **data leakage**. The steps, in order, are:

1. **Feature selection.** Keep the top n features ranked by the importance of a preliminary XGBoost model (200 trees, depth 3, scale_pos_weight set to the class ratio). Unlike ranking by correlation, this captures non-linear effects and handles the duplicated features found in Section 2.3. The number n is tuned between 10 and 40.
2. **IQR outlier removal.** Rows outside [Q1 − 1.5·IQR, Q3 + 1.5·IQR] on any selected feature are removed, using fences computed from the training data. Only **non-bankrupt** rows are removed. Applying the rule to every row on all 95 features would delete about 93% of the data and 219 of the 220 bankrupt companies, because bankrupt companies are outliers far more often than healthy ones.
3. **SMOTE (Synthetic Minority Over-sampling Technique).** New synthetic bankrupt examples are created by interpolating between existing ones until the two classes are balanced 1:1 in the training data.

4. Model Creation

XGBoost (Extreme Gradient Boosting) was chosen as the classifier. It is an ensemble of decision trees built one after another, where each new tree corrects the errors of the previous ones. It handles skewed, unscaled and correlated features well and captures non-linear relationships, which fits the characteristics found in the EDA. Two models were built and compared on the same split and the same folds.

Table 7. Models compared

Model	Description
Baseline	XGBoost with default hyperparameters, trained on all 95 features, with no outlier removal, no resampling and no tuning
Engineered	Feature selection + IQR outlier removal (non-bankrupt rows only) + SMOTE + tuned XGBoost

4.1 Baseline Model

The baseline is XGBClassifier with default settings (random_state = 42). It gives a reference point for how much the engineering steps help.

4.2 Engineered Model and Hyperparameter Tuning

Hyperparameters of the whole pipeline were tuned with **RandomizedSearchCV**: 30 random candidates × 10 folds = 300 fits. The search was scored by **PR-AUC** (average precision). PR-AUC does not depend on a decision threshold and focuses on the minority class, so it suits imbalanced data better than accuracy or ROC-AUC. Randomized search was preferred to grid search because it covers a large search space in far fewer runs.

Table 8. Hyperparameter search space and best values

Parameter	Search range	Best value
select__max_features (number of features)	10 – 40	40
clf__n_estimators	100 – 500	479
clf__learning_rate	0.01 – 0.2 (log scale)	0.0113
clf__max_depth	2 – 6	3
clf__min_child_weight	1 – 10	10
clf__subsample	0.6 – 1.0	0.643
clf__colsample_bytree	0.5 – 1.0	0.628

The best candidate reached a mean cross-validated PR-AUC of **0.426**. It chose the maximum allowed number of features (40), shallow trees (depth 3), a small learning rate with many trees, and strong regularisation (min_child_weight = 10 with row and column subsampling). This suggests the model needed to be kept simple to avoid overfitting the synthetic SMOTE samples.

4.3 Evaluation Metrics

- **Accuracy**: the share of all predictions that are correct. It is misleading on imbalanced data.
- **Recall** (sensitivity): the share of bankrupt companies the model catches. This is the most important metric for an early-warning system.
- **Precision**: of the companies flagged as bankrupt, the share that actually went bankrupt.
- **F1-score**: the harmonic mean of precision and recall.
- **PR-AUC** and **ROC-AUC**: threshold-free measures of how well the model ranks bankrupt companies above healthy ones.

Accuracy, recall, precision and F1 use the default decision threshold of 0.5.

5. Results

5.1 Cross-Validation Results (Training Set)

Table 9. 10-fold cross-validation results (mean ± SD)

Model	Accuracy	Recall	PR-AUC	ROC-AUC	Precision	F1
Baseline	0.972 ± 0.006	0.300 ± 0.099	0.498 ± 0.106	0.938 ± 0.022	0.642 ± 0.208	0.403 ± 0.121
Engineered	0.740 ± 0.027	0.948 ± 0.067	0.426 ± 0.115	0.928 ± 0.038	0.107 ± 0.014	0.192 ± 0.024

The engineered model's CV scores are slightly optimistic, because the same folds were used to choose its hyperparameters. The held-out test set therefore gives the fairer comparison.

5.2 Test Set Results

Table 10. Held-out test set results (1,364 companies, 44 bankrupt)

Model	Accuracy	Recall	PR-AUC	ROC-AUC	Precision	F1
Baseline	0.968	0.295	0.504	0.961	0.520	0.377
Engineered	0.746	0.977	0.403	0.944	0.111	0.199
Always "not bankrupt"	0.968	0.000	–	–	–	–

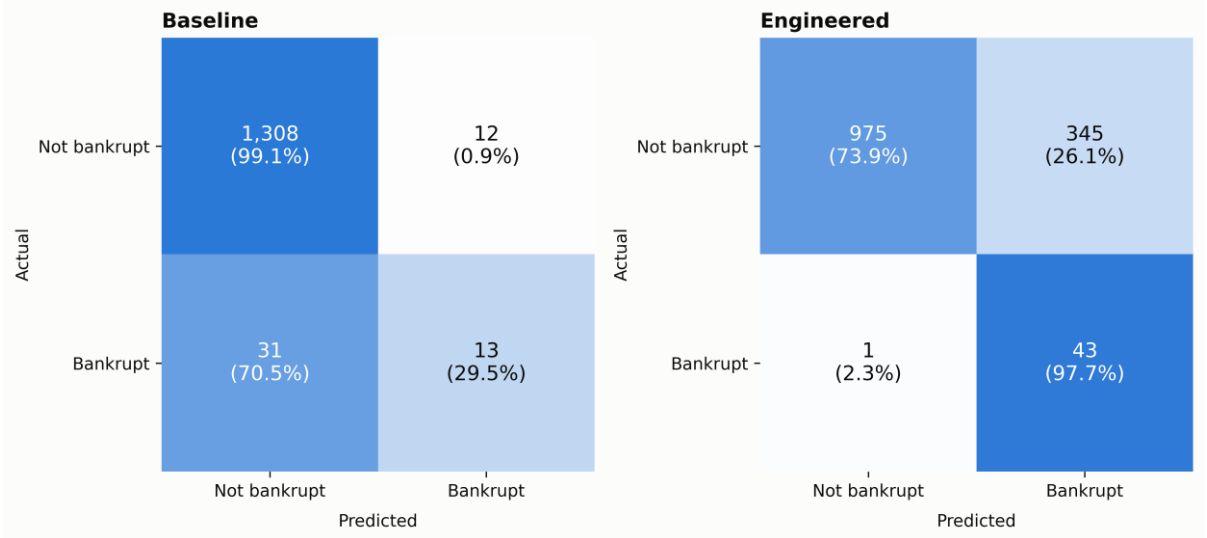


Figure 5.1. Confusion matrices on the test set (threshold 0.5)

Table 11. Confusion matrix counts on the test set

	Baseline	Engineered
True positives (bankrupt caught)	13	43
False negatives (bankrupt missed)	31	1
False positives (false alarms)	12	345
True negatives	1,308	975

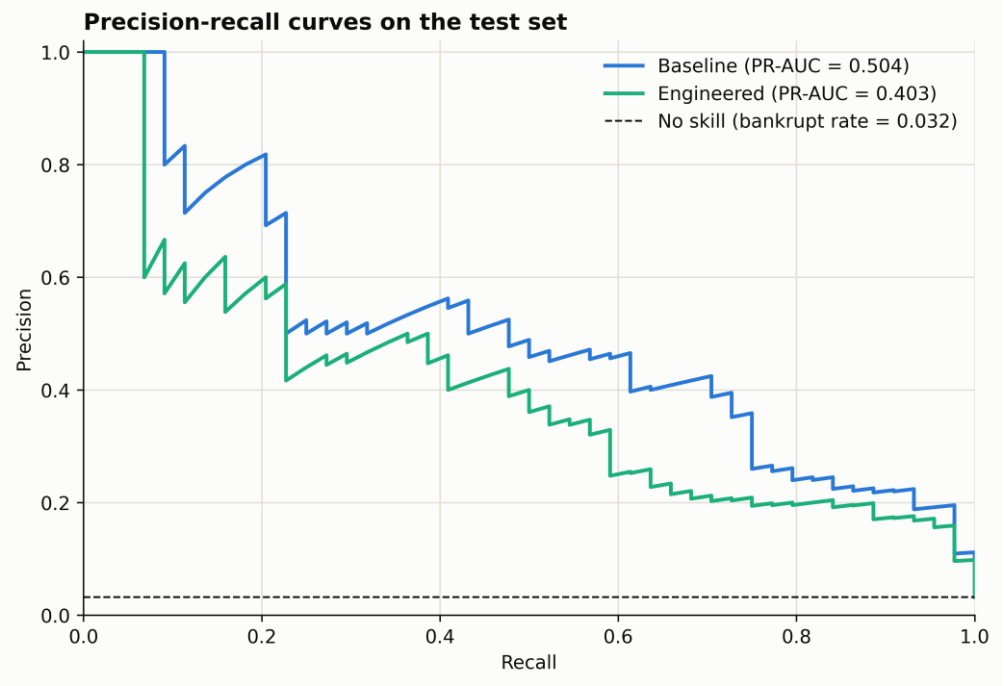


Figure 5.2. Precision–recall curves on the test set

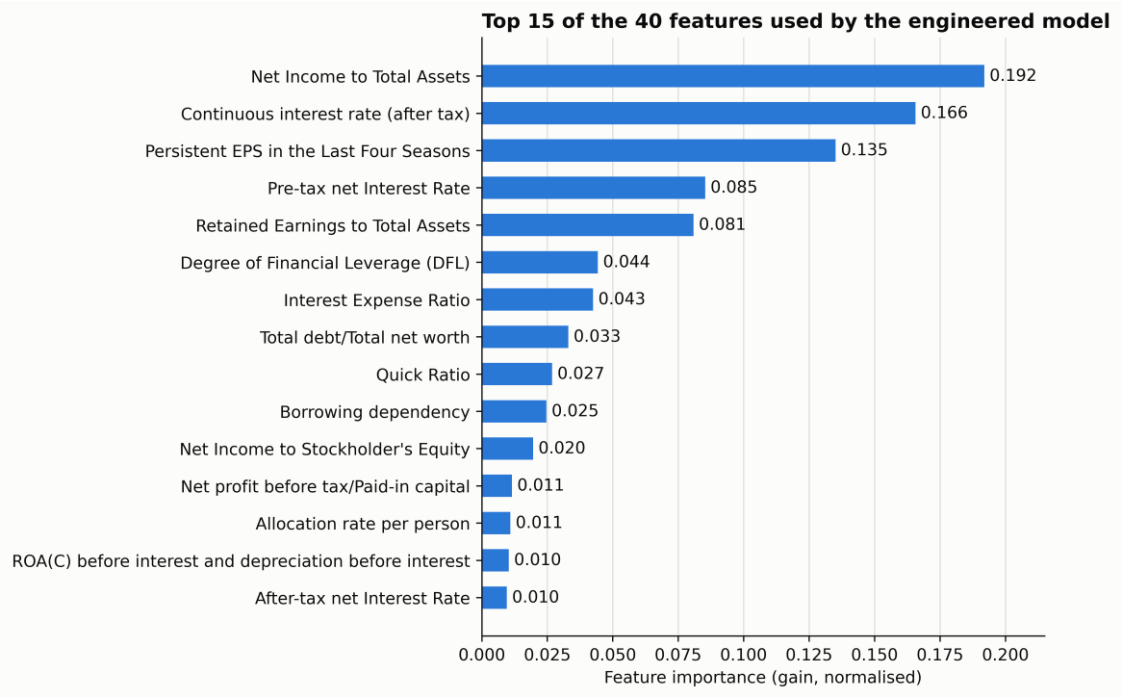


Figure 5.3. Top 15 of the 40 features used by the engineered model (gain importance)

The most important features of the engineered model are Net Income to Total Assets (0.192), Continuous interest rate after tax (0.166) and Persistent EPS in the Last Four Seasons (0.135), followed by Pre-tax net Interest Rate and Retained Earnings to Total Assets. Profitability and earnings dominate, which matches the correlation analysis in Section 2.2.

5.3 Discussion

Accuracy alone is misleading. The baseline's test accuracy (0.968) is exactly the same as a rule that never predicts bankruptcy. Its recall shows the real picture: it catches only 13 of the 44 bankrupt companies (29.5%) and misses 31.

The engineered model catches almost every bankrupt company. Its recall rises to 0.977: 43 of 44 bankrupt companies are caught and only 1 is missed. The price is accuracy, which falls to 0.746, because it flags 388 companies (about 28% of the test set) as bankrupt, 345 of them wrongly. Its precision is 0.111, meaning about 8 false alarms for every bankrupt company it catches.

Which model is better depends on the cost of each error. In bankruptcy prediction, a missed bankruptcy (false negative) usually costs a lender or investor much more than a false alarm (false positive), which only triggers an extra review by an analyst. From this point of view, the engineered model is more practical as an **early-warning screening tool**: it reduces the list of companies to check from 1,364 to 388 while missing only one bankruptcy. If false alarms are expensive, the baseline is preferable.

The engineered model does not rank companies better. The threshold-free PR-AUC is higher for the baseline (0.504 against 0.403 on the test set, and 0.498 against 0.426 in cross-validation). ROC-AUC is similar (0.961 against 0.944). The engineered model's high recall therefore comes mainly from flagging more companies at the 0.5 threshold, not from a better ability to separate the classes. Two steps cause this shift:

- IQR outlier removal takes extreme values out of healthy companies only, so the model learns that "an extreme value means bankruptcy".
- SMOTE balances the classes 1:1, which pushes predicted probabilities upward.

A similar trade-off could also be reached by lowering the decision threshold of the baseline model, without the extra engineering steps.

5.4 Limitations

- The test set contains only 44 bankrupt companies, so each company changes recall by about 0.023 and the results should be read with caution.
- All metrics except PR-AUC and ROC-AUC use a fixed threshold of 0.5. The threshold was not tuned to a business cost.
- Feature selection chose the upper limit of the search range (40 features), so allowing more features might help.
- Only one random split (random_state = 42) was used for the final test.

6. Conclusion

The dataset of 6,819 companies and 95 financial features is complete but highly imbalanced (3.23% bankrupt), heavily skewed, full of outliers and strongly multicollinear. The EDA showed that low profitability and high leverage are the clearest signs of bankruptcy, although no single feature separates the classes.

Two XGBoost models were compared. The baseline reaches 96.8% accuracy but catches only 29.5% of bankrupt companies, which is no better than predicting "not bankrupt" for everyone in terms of accuracy. The engineered model (feature selection, one-sided IQR outlier removal, SMOTE and randomized hyperparameter search) has lower accuracy (74.6%) but catches **97.7%** of bankrupt companies. This makes it the more useful model for identifying companies at risk of bankruptcy, as long as the extra false alarms can be handled by manual review.

The baseline still has the higher PR-AUC, so its ranking of companies is at least as good. Recommended next steps are to tune the decision threshold of both models against the real cost of missed bankruptcies and false alarms, to report an F2-score that weights recall above precision, and to confirm the results with repeated train–test splits.